ASSIGNMENT-1

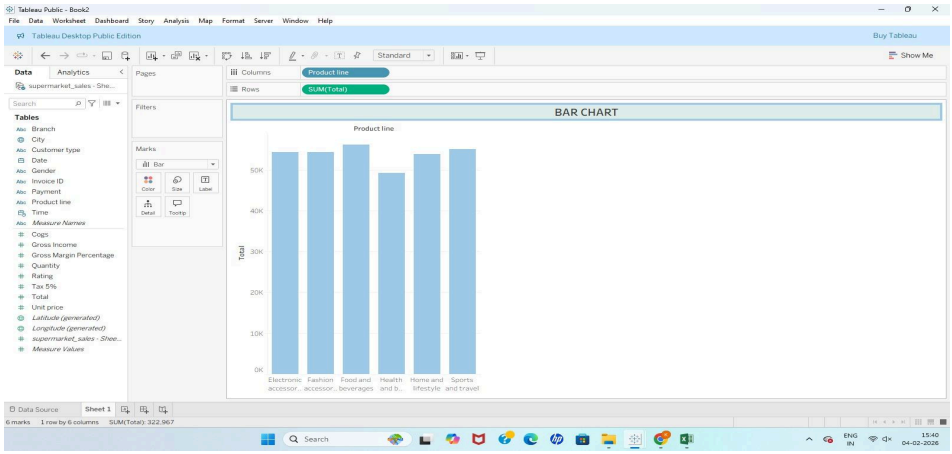
BAR CHART

Purpose::

A bar chart is primarily used to compare discrete quantities across different categories,track changes over time,and display data distributions. They are effective for visualizing nominal data (e.g., sales by product) and ordinal data (e.g., survey responses), making complex data sets easy to interpret

Process:

- Open Tableau Desktop Public Edition.
- Connect your dataset → supermarket_sales.
- Create a new worksheet.
- Drag Product line → Columns.
- Drag Total → Rows.
- Tableau automatically creates bars.
- In Marks:
- Select Bar.
- Optional formatting:
- Sort descending.
- Add labels (drag Total → Label).
- Title → BAR CHART.
- Result: Sales comparison for each product line.



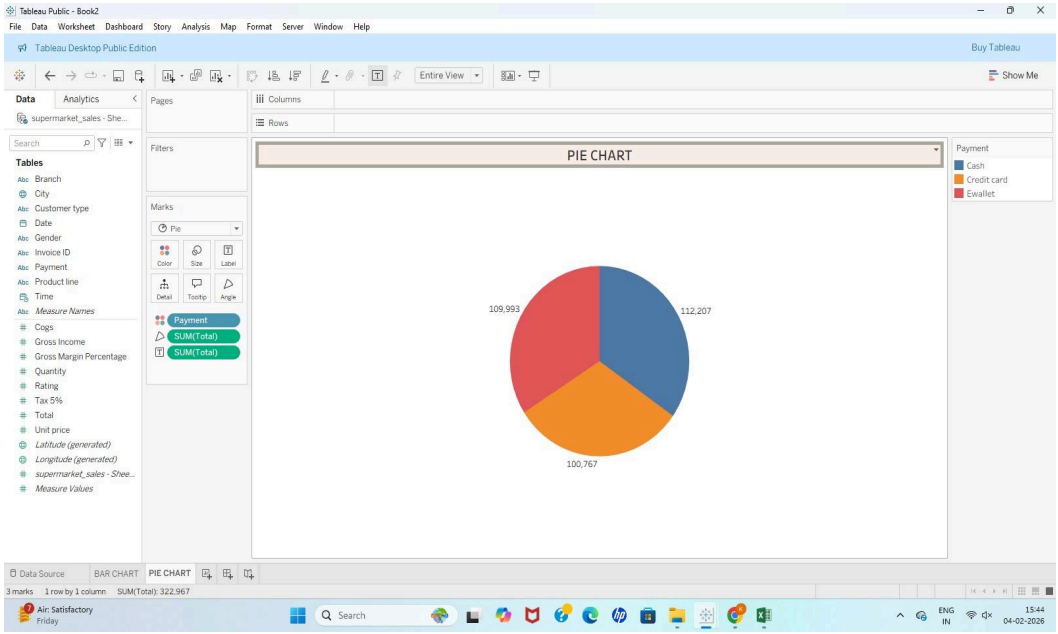
PIE CHART

Purpose:

A pie chart helps organize and show data as a percentage of a whole. True to the name, this kind of visualization uses a circle to represent the whole, and slices of that circle, or "pie", to represent the specific categories that compose the whole.

Process:

- Open a new worksheet.
- Change Marks type:
- Click dropdown → select Pie.
- Drag Payment → Color.
- Drag Total → Angle.
- Drag Total again → Label.
- Click Label → Show mark labels.
- Adjust pie size using Size slider.
- Title → PIE CHART.
- Result: Compare sales distribution by payment type.



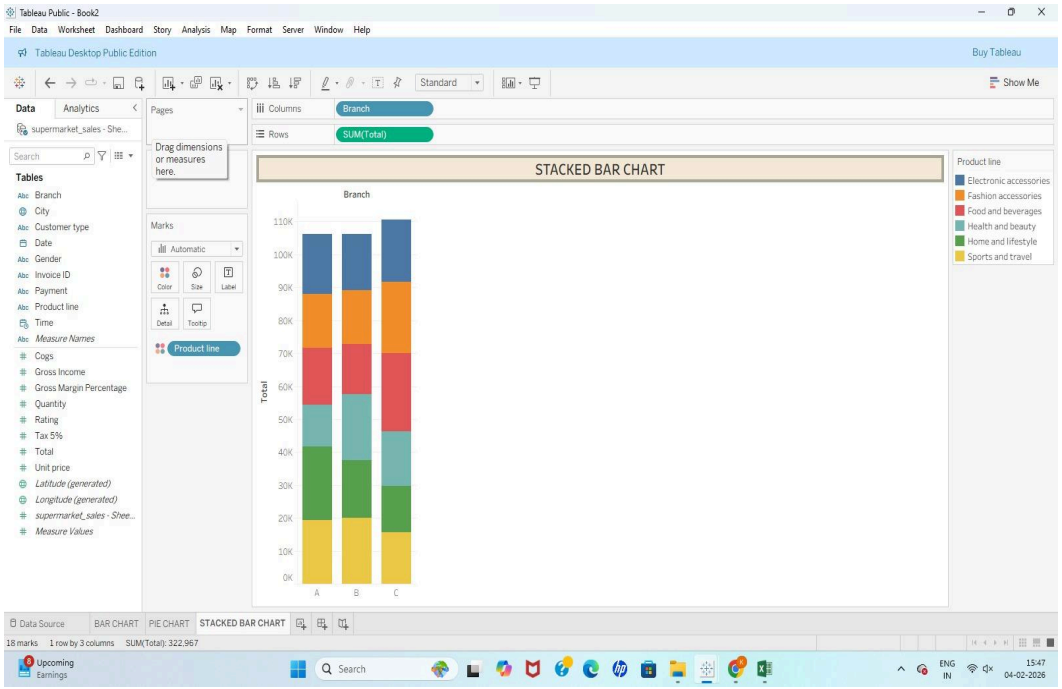
STACKED BAR CHART

Purpose:

A stacked bar chart, also known as a stacked bar graph, is a graph that is used to break down and compare parts of a whole. Each bar in the chart represents a whole, and segments in the bar represent different parts or categories of that whole.

Process:

- Open a new worksheet.
- Drag Branch → Columns.
- Drag Total → Rows.
- Drag Product line → Color (Marks card).
- Ensure stacked bars are enabled:
- Analysis → Stack Marks → On
- Chart type → Bar (Automatic works).
- Rename title → STACKED BAR CHART.
- Result: Each branch's sales broken into product categories.



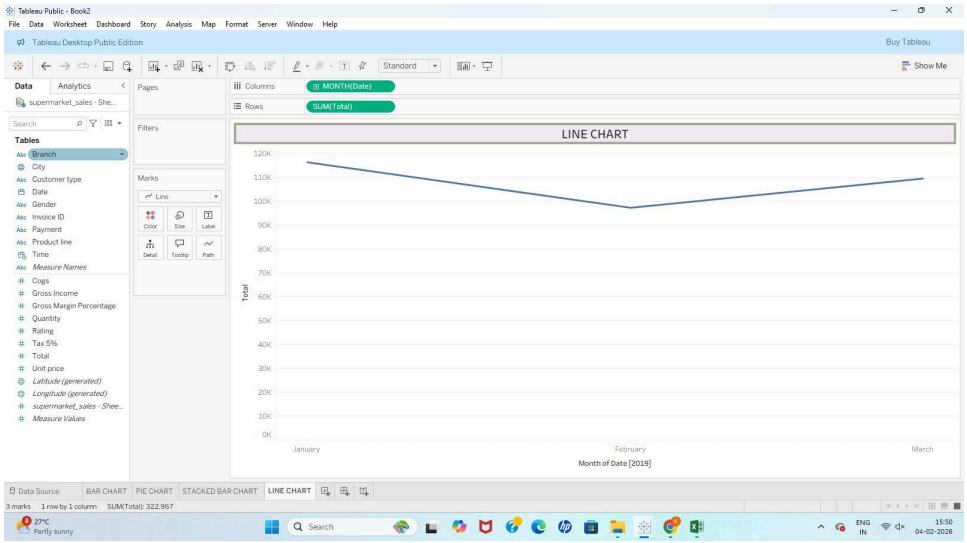
LINE CHART

Purpose:

A line chart, also referred to as a line graph or a line plot, connects a series of data points using a line. This chart type presents sequential values to help you identify trends. Most of the time, the x-axis (horizontal axis) represents a sequential progression of values.

Process:

- Open Tableau Desktop Public Edition.
- Connect your dataset → supermarket_sales.
- Go to a new worksheet.
- Drag Date → Columns shelf.
- Click the Date pill → choose MONTH(Date).
- Drag Total → Rows shelf.
- Tableau automatically creates a line chart.
- In Marks, select:
- Chart type → Line
- Format:
- Rename title → LINE CHART
- Adjust view → Standard
- Result: Monthly sales trend (January–March).



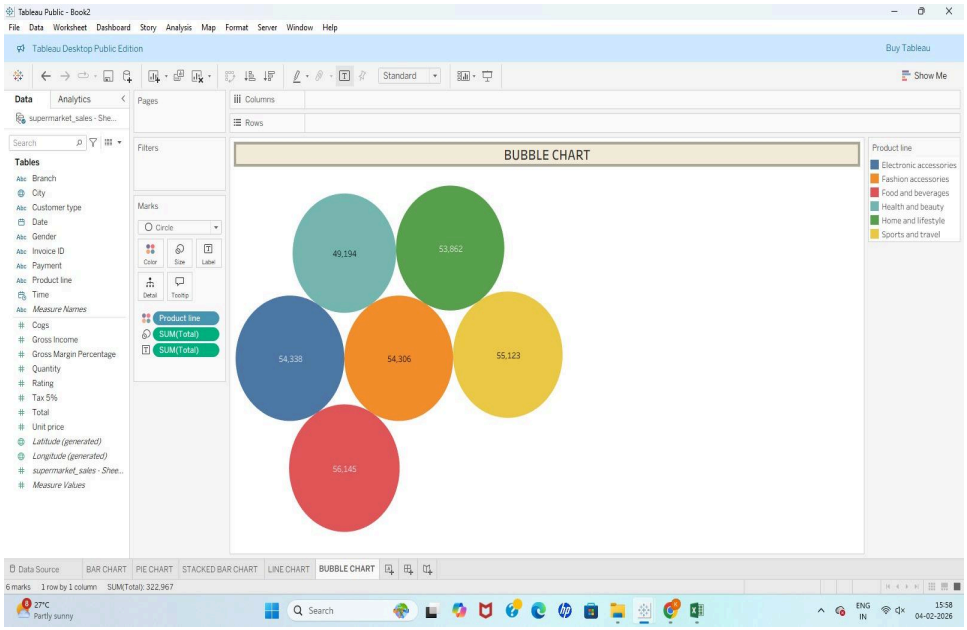
BUBBLE CHART

Purpose:

A bubble chart is a data visualization tool that displays three to four dimensions of data simultaneously, using x and y coordinates for two variables, bubble size for a third, and color for an optional fourth. It is primarily used to identify relationships, trends, and patterns among large datasets.

Process:

- New worksheet → Rename Bubble Chart
- Change Marks type to Circle
- Drag Product line → Color
- Drag Total → Size
- Drag Total → Label
- Click Show Me (top right) → Select Packed Bubbles
- Adjust Size slider
- Edit Title → BUBBLE CHART
- Result:Compare product line sales.



ASSIGNMENT-2

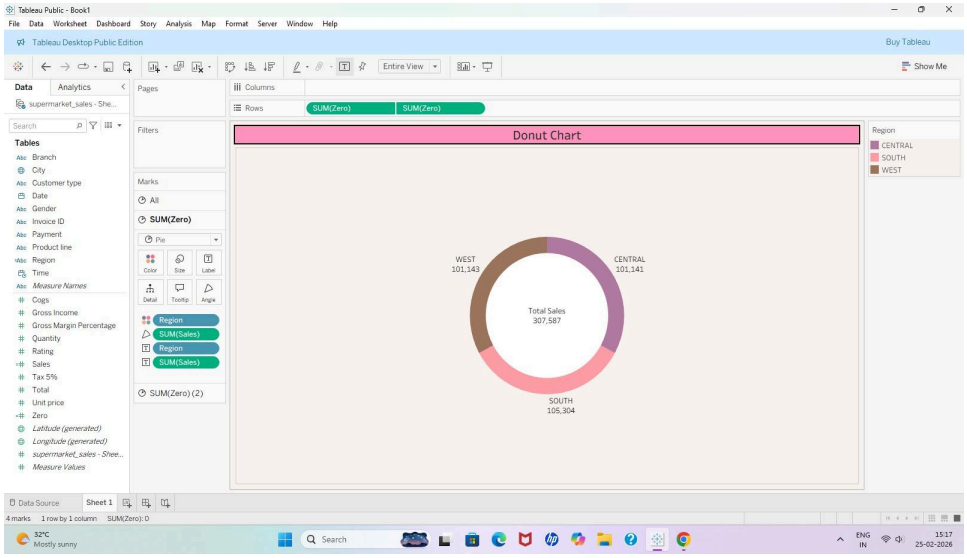
DONUT CHART

Purpose:

A donut chart is best used to represent parts of a whole, but this is not to be confused with adding up to the number hundred. Any value, even one that runs into a few million, can be represented as a donut chart, as long as all of the values in total represent parts of a whole or 100 percent.

Process:

- Open Tableau Public/Desktop and load your dataset.
- Click on a new Sheet.
- Go to Show Me → Pie Chart (or change Marks type to Pie).
- Drag Category → to Color ,Drag Category → to Label
- Drag Quantity (or Sales field) → to Angle
- Drag it again → to Label (to show % or values)
- Drag the same measure to Rows twice
- Right-click second axis → Dual Axis
- Change second Marks to Circle
- Reduce Size and color it white (to create donut hole)
- Title:DONUT CHART



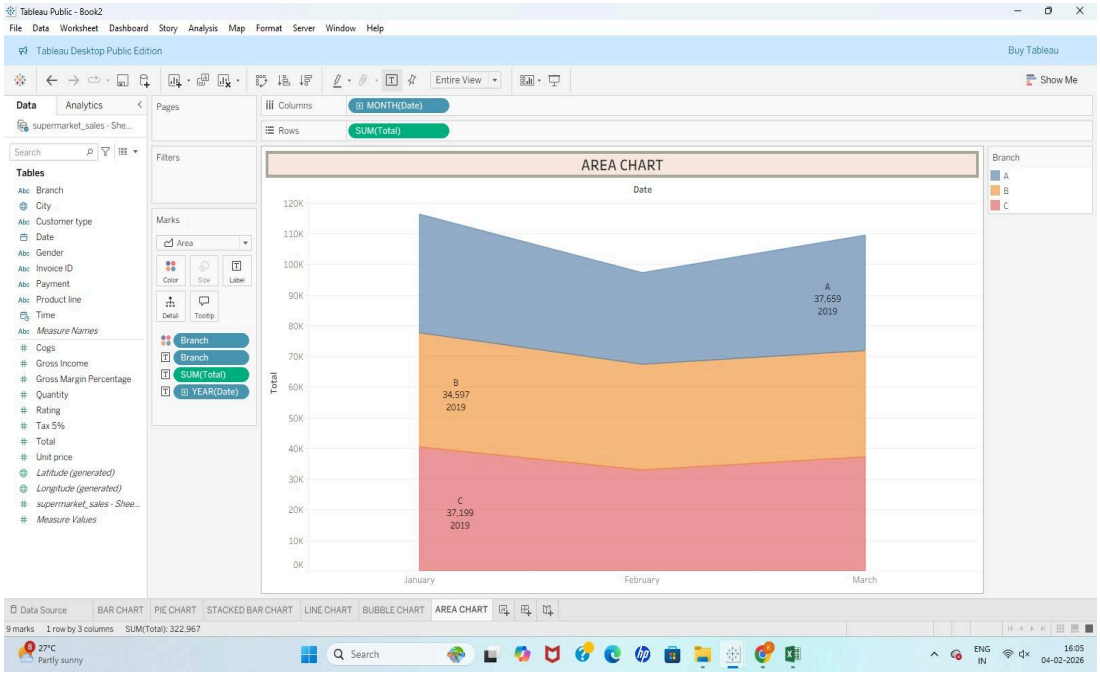
AREA CHART

Purpose:

An area chart is a visual representation of data that utilizes both lines and filled areas to convey information. This type of chart is particularly effective in showcasing data trends and variations over a specified period or across different categories.

Process:

- Open Tableau and load your dataset.
- Click on a new sheet.
- Drag Date → to Columns.
- Change it to MONTH(Date).
- Drag Total → to Rows (ensure it shows SUM(Total)).
- In Marks card → choose Area
- Drag Branch → to Color (to create stacked areas).
- Rename the title to: AREA CHART



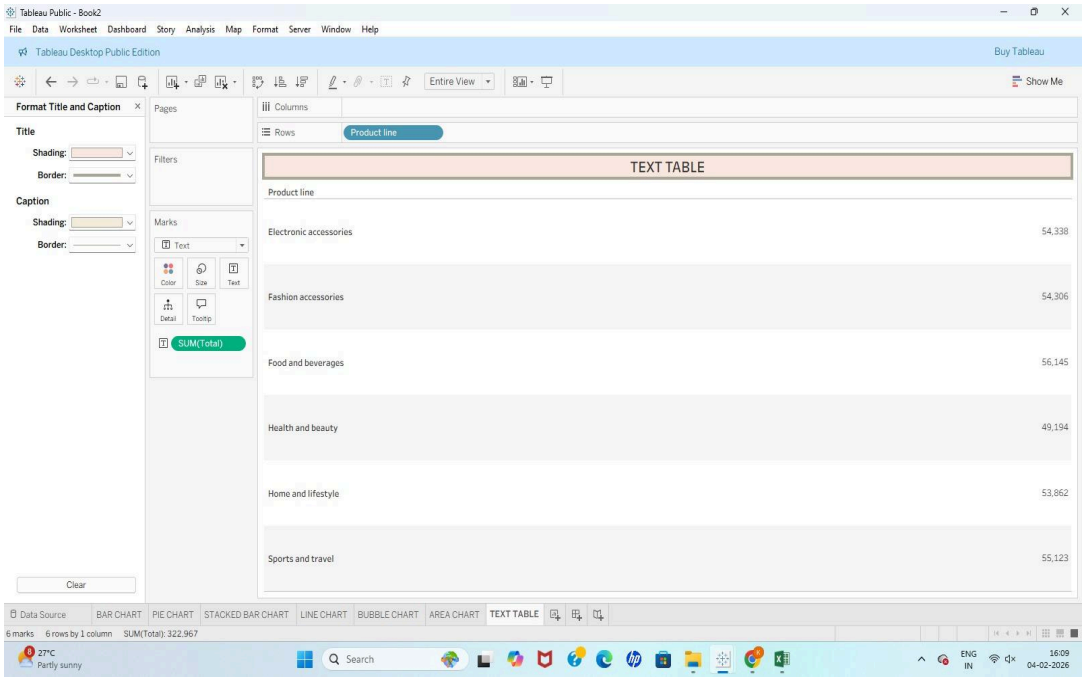
TEXT TABLES

Purpose:

Tables are used to organize data that is too detailed or complicated to be described adequately in the text, allowing the reader to quickly see the results. They can be used to highlight trends or patterns in the data and to make a manuscript more readable by removing numeric data from the text.

Process:

- New worksheet → Rename Text Table
- Drag Product line → Rows
- Drag Total → Text
- Format:
- Increase font size
- Add title TEXT TABLE
- Result:total sales by product line.



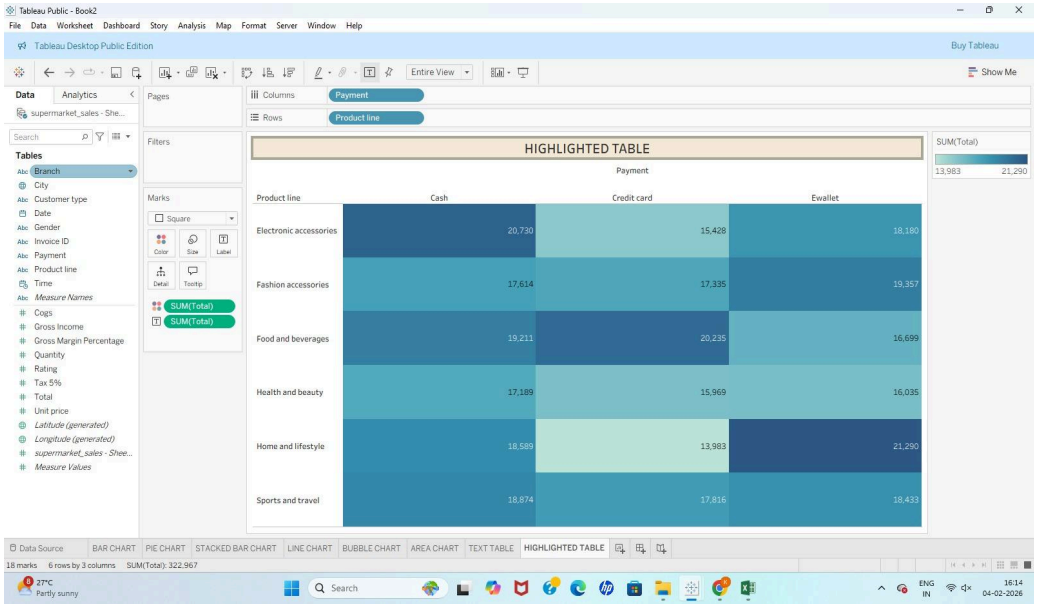
HIGHLIGHTED TABLE

Purpose:

A Highlight Table uses color coding within a table format to draw attention to specific data points, making it easier to identify important values and trends at a glance.Highlight tables are a data visualization tool combining text tables with color-coding to make large datasets instantly interpretable.

Process:

- Open new sheet → Rename Highlighted Table
- Drag Payment → Columns
- Drag Product line → Rows
- Drag Total → Marks → Color
- Drag Total again → Marks → Label
- Change Marks type to Square
- Click Color → Edit Colors
- Choose a single color gradient (like blue)
- Increase Size slider so squares fill space.
- Edit Title → HIGHLIGHTED TABLE
- Result:Compare Total Sales by Product line & Payment Type



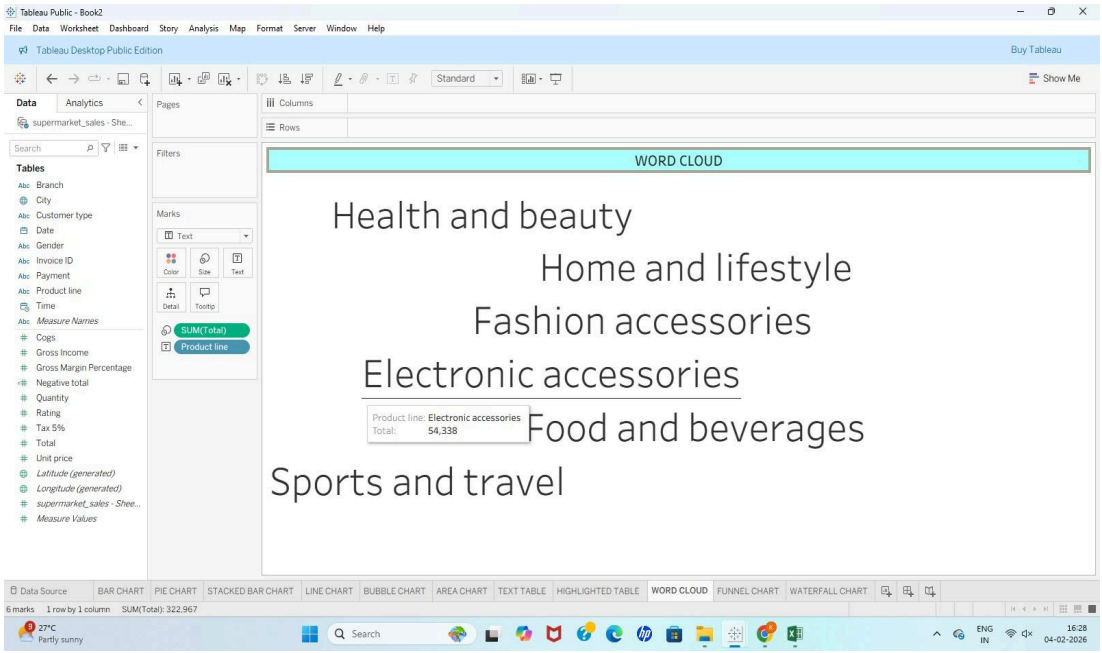
WORD CLOUD

Purpose:

The word cloud is a visual tool for displaying the most frequently used words in a text or set of texts. These words are displayed in different sizes, colours and orientations, depending on how often they are used. The more often a word is used, the larger it appears in the cloud.

Process:

- New worksheet → Rename Word Cloud
- Change Marks type to Text
- Drag Product line → Text
- Drag Total → Size
- Drag Total → Label
- Increase Size slider
- Edit Title → WORD CLOUD
- Result:Show Product lines sized by total sales



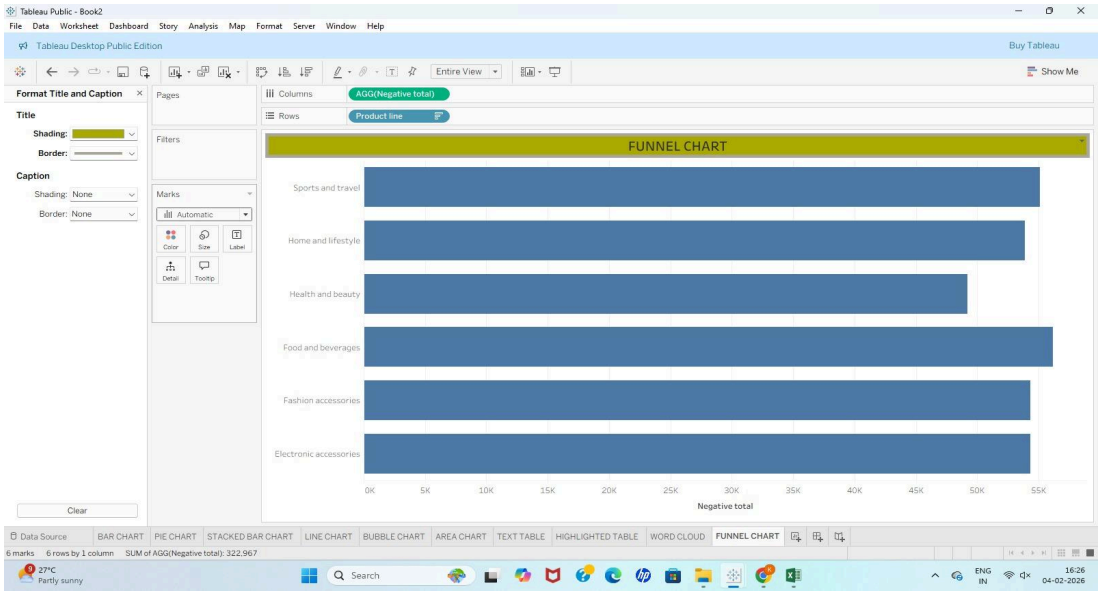
FUNNEL CHART

Purpose:

Funnel analysis provides a visual representation of a critical user journey in your product. The horizontal axis represents each step that a user must pass through. The vertical axis represents the percentage of users or sessions at each step.

Process:

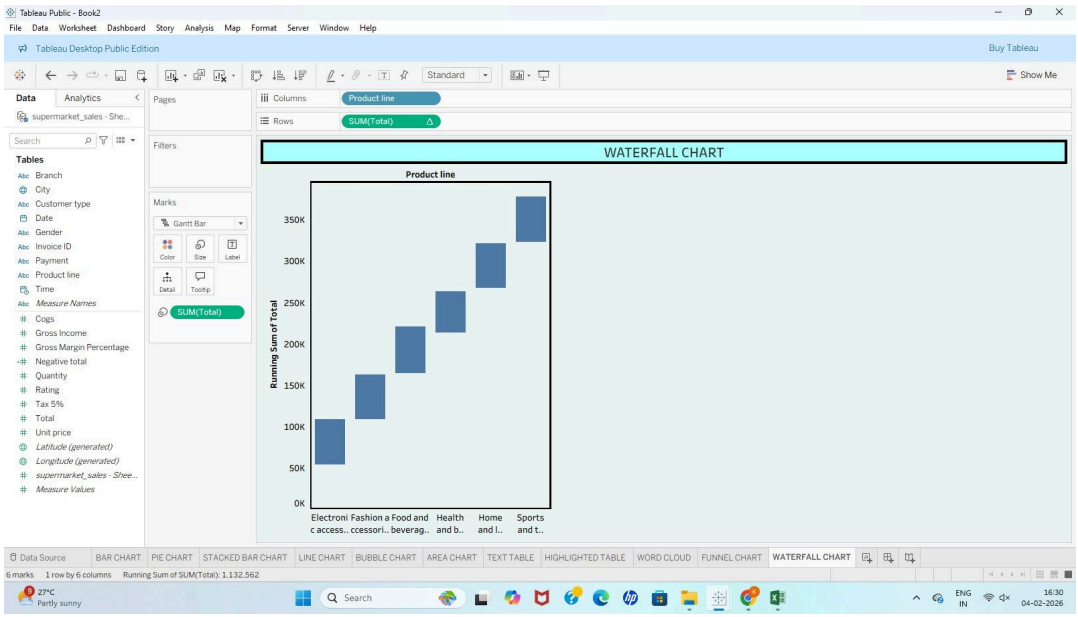
- Create Calculated Field:
- Name: Negative Total
- Formula:
- -SUM([Total])
- New worksheet → Rename Funnel Chart
- Drag Negative Total → Columns
- Drag Product line → Rows
- Change Marks to Bar
- Sort descending
- Format axis:
- Right-click axis → Edit Axis
- Remove negative sign display if needed
- Edit Title → FUNNEL CHART



WATERFALL CHART

Purpose:

A waterfall chart shows a running total as values are added or subtracted. It's useful for understanding how an initial value (for example, net income) is affected by a series of positive and negative values. The columns are color coded so you can quickly tell positive from negative numbers.

Process:

- Open a new worksheet → Rename it Waterfall Chart.
- Drag Product line → Columns
- Drag Total → Rows
- Right-click on SUM(Total) in Rows
- → Select Quick Table Calculation
- → Choose Running Total
- Change Marks type to Gantt Bar
- Drag SUM(Total) again → Drop on Size
- (Optional)
- Edit Title → Type WATERFALL CHART